

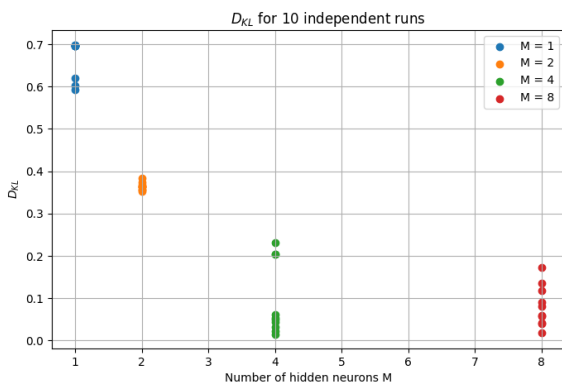
In this assignment we train a restricted Boltzmann machine with 3 visible neurons and the number of hidden neurons taking on different values $M = 1, 2, 4, 8$, where the resulting network is used in order to model the XOR distribution. All neurons take on states either 1 or -1 and the connecting weights $w_{ij}, i = M, j = 3$ were initialized as samples from the standard normal distribution while the thresholds for both the visible and hidden states were set as 0 ($\theta_j^{(v)} = \theta_i^{(h)} = 0$).

The training was done using the CD- k algorithm for ± 1 neurons, specifically the one described in [1] (Algorithm 3), with parameters $\nu_{max} = 10^5, p_0 = 10, k = 10$ and learning rate $\eta = 0.005$. After training, the learned Boltzmann distribution was estimated by running the dynamics of the restricted Boltzmann machine until reaching a steady state. This was done by using 10^4 burn-in iterations followed by 10^6 samples to evaluate the probabilities. The resulting count of the patterns were used to calculate the Kullback-Leibler divergence D_{KL} from the target XOR distribution. For each value of M , 10 independent training runs were performed and the results were compared to that of the theoretical upper bound (Eq. 4.40 in [1]).

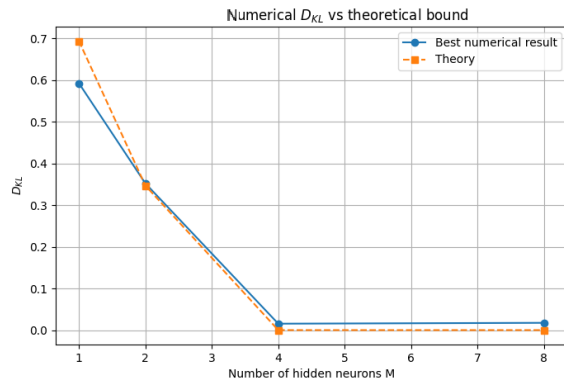
In table 1 and figure 1 we observe that when we increase the number of hidden neurons M , D_{KL} decreases substantially. The best values from the ten independent runs decreased from 0.592 for $M = 1$ to 0.352 for $M = 2$, then for $M = 4$ and $M = 8$ the values were 0.015 and 0.017 respectively. As such, for $M = 4$ and $M = 8$ the estimated values were already close to 0 and showed little improvement.

$M = 1$	$M = 2$	$M = 4$	$M = 8$
0.592	0.352	0.015	0.017

Table 1: Best D_{KL} obtained from 10 independent runs for each number of hidden neurons $M \in \{1, 2, 4, 8\}$.



(a) All 10 independent D_{KL} values versus M .



(b) Best numerical D_{KL} and theoretical bound versus M .

Figure 1: Results of the restricted Boltzmann machine trained on the XOR data set.

The decrease in D_{KL} with increasing M is consistent with the increased capacity of the restricted Boltzmann machine. For $N = 3$ visible neurons, the theoretical upper bound is 0 for $M \geq 3$, meaning that the XOR distribution can be represented with arbitrary small D_{KL} . This supports the low values found for $M = 4$ and $M = 8$. The remaining non-zero D_{KL} for $M = 4$ and $M = 8$ may be explained by the usage of the CD- k algorithm, which is not guaranteed to converge to the optimal solution [1].

Overall, the restricted Boltzmann machine successfully learned the XOR distribution, with substantial shown improvements between $M = 1$, $M = 2$ and $M = 4$ hidden neurons and no great difference between $M = 4$ and $M = 8$ hidden neurons. All of which being consistent with the theoretical result.